

David Shu-Yu Wu

0902-178-182 | b12902036@ntu.edu.tw | [LinkedIn](#) | [GitHub](#) | [Website](#)

EDUCATION

National Taiwan University

B.S. in Computer Science and Information Engineering; Minor in Atmospheric Sciences

Taipei, Taiwan

Sep. 2023 – Present

PROFILE

CS senior at National Taiwan University focused on cryptographic engineering and post-quantum cryptography. Experienced in NTT-based big-integer multiplication, modular arithmetic, constant-time cryptographic implementation, and ARM assembly optimization. Additional hands-on exposure to formal verification and equivalence checking for optimized cryptographic kernels and implementation-security topics such as synthetic trace simulation for side-channel analysis. Broader academic interests include theoretical computer science, algorithms, and probabilistic methods.

RESEARCH EXPERIENCE

Fast Crypto Lab

Jul. 2025 – Present

Research Intern, Academia Sinica, supervised by Bo-Yin Yang

Taipei, Taiwan

Multiplying Not-So-Big Integers with FFTs – Finding the Crossover Point on Large Modern OoO ARM CPUs

- Implemented NEON-optimized Number-Theoretic Transform (NTT) kernels and modular reduction (Barrett, Montgomery) in AArch64 assembly on ARM Cortex-A76 (Raspberry Pi 5) for big integer multiplication.
- Built a reproducible benchmarking and unit-testing harness comparing our NTT-based multipliers against GMP across multiple operand sizes (e.g., 8k–14k bits), profiling cycle counts.
- Integrated an NTT-based multiplier into OpenSSL RSA and benchmarked RSA-8192/10240 operations, improving public-key operations such as verification, encryption, and KEM encapsulation by 36.0%–44.7%.
- Formally verified optimized large integer multiplication subroutines using CryptoLine and HOL Light, reasoning about algebraic correctness, range bounds, and equivalence between reference and Slothy-optimized implementations.

NTU Global Connect Fellowship Program

Jul. 2026 – Aug. 2026

Research Intern, EEE, Nanyang Technological University, supervised by Gwee Bah Hwee

Singapore

Generative AI for High-Fidelity Side-Channel Trace Simulation

- Selected for the NTU Global Connect Fellowship, a competitive international summer research program with an acceptance rate below 2%.
- Reviewing profiling side-channel analysis literature, including masking, desynchronization, leakage modeling, data augmentation, and synthetic trace generation for cryptographic implementations.
- Studying how synthetic side-channel traces can be used to augment profiling datasets and evaluate attack robustness in machine-learning-based side-channel analysis.

COURSE PROJECTS

Cryptographic Engineering: ML-KEM & X25519 on Cortex-M4 | C, ARMv7E-M ASM

Fall 2025

- Optimized constant-time X25519 (ECDH25519) on Cortex-M4, including variable-base scalar multiplication and fixed-base window method (constant-time table scan + conditional move).
- Reworked field arithmetic modulo $2^{255} - 19$: implemented mixed-radix (25/26-bit limbs) multiplication with product-scanning, and specialized squaring to eliminate redundant cross terms; translated hotspots to ARMv7E-M assembly to reduce cycle counts.
- Optimized ML-KEM polynomial arithmetic: implemented NTT/InvNTT and base multiplication in assembly with layer fusing and controlled unrolling; leveraged ARMv7E-M DSP packed 16-bit operations for poly add/sub under a strict per-routine NTT code-size budget.
- Achieved up to **94%** cycle-count reduction for X25519 fixed-base scalar multiplication and **62%** reduction for ML-KEM poly_ntt vs. course baseline under the same benchmarking setup.

Fuzzy Counting with Probabilistic Data Structures | C++, Python

Spring 2025

- Implemented typo-tolerant fuzzy matchers using Bloom filter segmentation, Cuckoo filters, HyperLogLog, and HyperLogLog-LDP.
- Evaluated accuracy–space tradeoffs over synthetic typo datasets under different sketch parameters and segmentation strategies, exposing the implementation as a reusable Python library.

Operating Systems Projects | C

Spring 2025

- Extended xv6 with user-level threads, cooperative context switching, and a SLAB allocator with freelist management and partial/full slab classification.
- Implemented and evaluated CPU schedulers (HRRN, P-RR, DM, EDF-CBS) and simulated RAID-1 fault-tolerant file operations in a Docker-based development environment.

CAPTCHA Generation and Evaluation Platform | *Python*

Spring 2025

- Built a modular CAPTCHA evaluation platform with perturbation-based image generation, ML-based attack testing, and a Streamlit interface.

COURSES

Relevant Coursework: Cryptography and Network Security, Post-Quantum Cryptography, Cryptographic Engineering, Cryptanalysis, Algorithm Design and Analysis, Advanced Data Structures, Probability, Discrete Mathematics, Automata and Formal Languages, Information Theory and Coding Techniques, Operating Systems, Computer Architecture, Machine Learning, Data Structures and Algorithms, System Programming, Linear Algebra, Computer Networks

TECHNICAL SKILLS

Languages: C, C++, Python, Java, AArch64 assembly, ARMv7E-M assembly

Cryptography / Verification: ARM NEON intrinsics, CryptoLine, GMP, OpenSSL

Tools & Platforms: Git, Docker, Linux, Raspberry Pi 5, VS Code